

REA: RECURSIVE ENGINEERING AGENTS WITH VERSIONED TOOL, PARTS, AND DATASET MEMORY

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ABSTRACT

Engineering agents must combine procedural knowledge with evidence about tools, components, datasets, and their compatibility. This working draft presents Recursive Engineering Agents (REA), an experimental adaptation of procedural scientific skills to recursive language model execution over versioned external memory. REA separates skills, tools, parts, datasets, and supporting evidence, and provides specialist profiles covering 71 engineering domains and 609 sub-categories. Explicit category membership scopes catalog access while preserving source references. The current implementation supports bounded recursive execution and immutable record addressing; it supports recursive, multi-turn analysis of external corpora larger than an individual model context window, with practical capacity governed by memory and computational budgets. We report implementation checks and a catalog census, and specify a comparative evaluation protocol. Engineering task accuracy, gains over retrieval baselines, and generalization across domains remain unmeasured. This draft is a research scaffold and is not ready for submission.

1 INTRODUCTION

Engineering reasoning depends on more than identifying a plausible procedure. An agent may need to determine whether a solver supports a boundary condition, whether a component meets an interface constraint, or whether a dataset contains the required variables. The relevant evidence may span procedural documents, tool descriptions, component catalogs, and dataset documentation. Providing this material as a single model prompt couples the accessible evidence to the model’s finite input capacity.

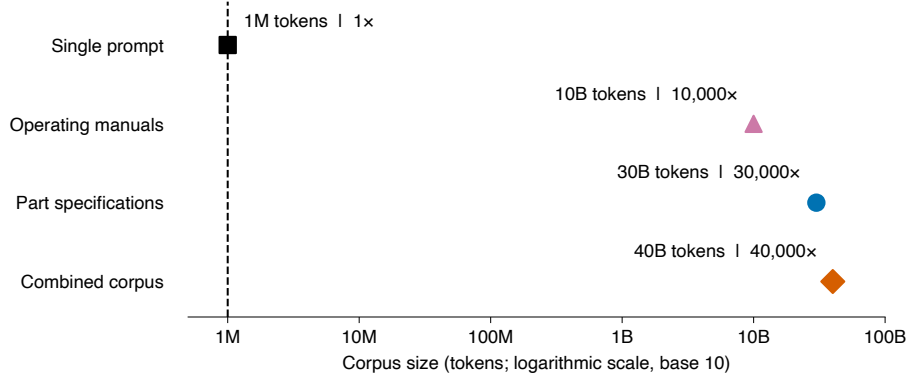
Scientific Agent Skills organizes procedural knowledge for research agents (Kassis et al., 2026). Recursive Language Models expose long inputs as an external environment that can be inspected programmatically and processed through recursive model calls (Zhang et al., 2025). REA combines these mechanisms with versioned engineering catalog records. Its proposed contribution is a reproducible way to scope and trace engineering evidence across recursive investigations. Whether this combination improves engineering outcomes is the central empirical question, not an established result.

The implementation contribution consists of separate memory namespaces, stable record identity, catalog provenance checks, and a complete engineering taxonomy shared between runtime specialist profiles and this appendix. The planned scientific contribution requires demonstrating that these mechanisms improve independently checked task outcomes under matched resource budgets. We do not infer capability from the number of indexed records or the number of specialist profiles.

2 CORPUS-SCALE MOTIVATION

Figure 1 illustrates the scale difference between a fixed prompt and externally accessible engineering evidence. Under its assumptions, even allocating an entire one-million-token window to specifications would fit at most 2,000 parts, or approximately 0.0033% of the catalog. Actual prompt capacity would be lower after instructions and other input are included. Selective external access allows an

Analytical scenario: 60 million parts + 1 million operating manuals



Explicit assumptions

Parts: 60M × 500 tokens = 30B. Manuals: 1M × 10,000 tokens = 10B.

Sensitivity: 100–2,000 tokens/part gives 6,000–120,000 prompt equivalents.

At 500 tokens/part, a 1M-token prompt holds at most 2,000 parts (0.0033%).

Capacity illustration, not measured REA throughput, accuracy, or exhaustive coverage.

Figure 1: Analytical corpus-scale scenario, not a benchmark result. A hypothetical 60-million-part catalog at 500 tokens per specification requires 30 billion tokens. A separate one-million-document operating-manual corpus at 10,000 tokens per document adds 10 billion. Together they represent 40,000 one-million-token prompt equivalents, or approximately 4.6 orders of magnitude. Token densities and counts are explicit assumptions; the parts sensitivity range is not a confidence interval. Prompt equivalents describe corpus size, not the number of model calls, coverage, speedup, or measured task performance.

agent to investigate candidates across turns without placing the entire catalog in a single call. It does not imply that every record is read or that all relevant records are found.

This capacity distinction is shared with retrieval-based systems. REA’s proposed additional benefit is recursive decomposition: subcalls can inspect component specifications and operating procedures separately, return cited intermediate findings, and trigger further investigation of unresolved constraints (Figure 2). Demonstrating that this mechanism improves accuracy or cost requires a matched multi-turn retrieval baseline. The 60-million-part scenario has not been ingested or evaluated in the current implementation; paged access and durable execution state are required engineering work for that scale.

3 RELATED WORK

Procedural skill libraries externalize instructions that can be loaded when relevant (Kassis et al., 2026). Recursive Language Models instead address how a model can inspect and decompose a large external input (Zhang et al., 2025). REA adopts the latter execution mechanism rather than introducing a new recursive model architecture. Persistent memory systems such as MemGPT address management of information beyond the immediate prompt (Packer et al., 2023); the distinctive engineering question here concerns versioned evidence, compatibility constraints, and domain scoping.

A scite-assisted discovery pass produced 262 distinct DOI candidate records from twelve queries and a capped citation-graph expansion. These are candidate literature records, not a complete systematic review. Screening and metadata reconciliation remain necessary before making novelty claims. The engineering comparison must cover domain-specific simulation agents, CAD and component retrieval, structural design, controls, and materials workflows. A catalog survey cannot substitute for direct comparisons with the most relevant systems.

4 METHOD

Let a memory record be $r = (n, i, v, b)$, where n is its namespace, i its identifier, v its version, and b its body. The store rejects changes to an existing namespace–identifier–version tuple. The five namespaces contain procedural skills, engineering tools, parts-source records, dataset-source records, and supporting evidence. The parts and dataset namespaces currently describe sources rather than complete inventories of components or datasets.

4.1 TAXONOMY AND SPECIALIST SCOPE

The taxonomy defines domains D and subcategories C_d for each domain d . A tool record has explicit catalog memberships $M(r)$. For a requested domain, the scoped tool set is

$$T_d = \{r \in T : M(r) \cap C_d \neq \emptyset\}. \quad (1)$$

Selecting one subcategory replaces C_d with that singleton. Unknown domains and categories belonging to another domain are rejected. Membership is editorial metadata; it does not certify task suitability, numerical accuracy, or availability of a tool.

Every domain receives a specialist role specification containing its complete subcategory definitions and instructions to preserve assumptions, units, versions, interfaces, uncertainty, and cited findings. These are profiles for recursive calls, not 71 independently trained or empirically qualified agents. Cross-domain work requires explicit handoffs. The runtime supplies the profiles to the recursive model; automatic optimal routing is not established.

The scope operation selects supporting evidence referenced by the retained tool records and shared parts and dataset records. Missing evidence identifiers are exposed. Skills and shared parts and dataset records remain available. Consequently, domain scoping reduces tool evidence but does not promise that every retained record is relevant to the task.

4.2 RECURSIVE EXECUTION AND RESOURCE BOUNDS

The experimental adapter uses the authors’ RLM implementation with an external Python dictionary in a Docker environment. The model can inspect this dictionary and request recursive subcalls. The default configuration caps depth at three, iterations at twenty, elapsed execution at 300 seconds, and tokens at 100,000, with four concurrent requests. A dollar threshold is a soft budget guard rather than a guarantee against an in-flight request exceeding the threshold.

The current adapter materializes the selected corpus in host memory before execution. A configurable serialized-character guard limits the exposed payload. It is therefore neither a paged storage service nor an unlimited context window. Each model call remains bounded by the underlying model context, and the system remains bounded by memory, latency, and compute. Scalability must be measured across corpus sizes and task families.

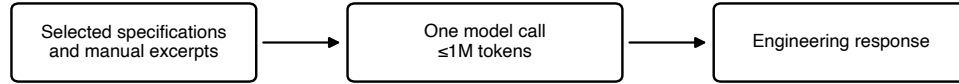
5 CURRENT IMPLEMENTATION EVIDENCE

The local snapshot contains 1,193 skill and reference documents, 4,795 tool records, 48,694 evidence records, 18 parts-source records, and 10 dataset-source records. The tool taxonomy contains 71 domains, 609 subcategories, and 5,658 tool–subcategory membership edges. Counts describe coverage of the imported snapshot, not unique engineering capabilities or validated workflows.

Fourteen deterministic extension tests and ten repository metadata tests passed. The extension tests include immutable addressing, import and retrieval behavior, complete taxonomy counts, scope membership, cross-domain rejection, and missing-evidence reporting. Six local catalog scope audits, covering CAD, optics, CFD, structures, digital electronics, and controls, found no missing referenced evidence IDs in the scoped records. This establishes reference availability for those audits, not factual correctness of the evidence.

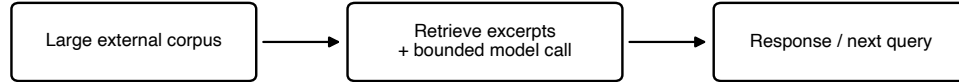
A previously recorded synthetic smoke test asked the recursive runtime to combine two load records containing 17 and 29. It returned 46 with both record identifiers. The trace recorded two recursive subcalls, five total model calls, 13,965 input tokens, and 13,517 output tokens over approximately

1 Single-call prompt



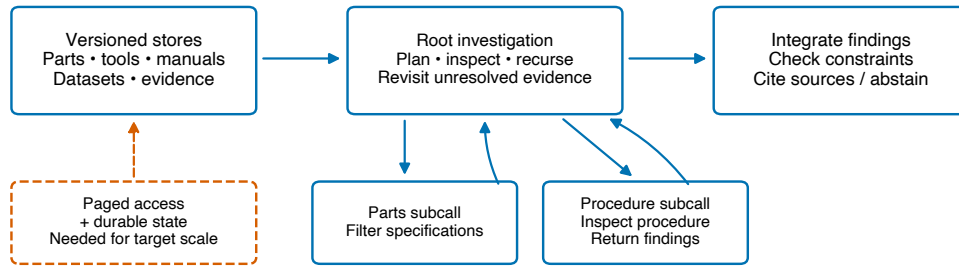
Only supplied evidence is visible in this call; selection must happen upstream.

2 Retrieval-based agent



Also accesses corpora larger than a prompt and can iterate across turns.

3 REA recursive investigation



Every model call remains bounded. Selective access does not imply reading every record.

Current REA materializes a bounded corpus in RAM. 60M-part operation is a design target.

Figure 2: Access patterns and proposed recursive advantage. A single model call sees only its supplied evidence. A retrieval-based agent can also access large corpora and iterate. REA supplies an external environment for inspection and recursive subcalls, enabling separate investigations of specifications and procedures followed by source-linked integration. All model calls remain bounded. The dashed paging and durable-state component is required for the target scale and is not implemented by the current RAM-materializing adapter. This schematic is not an execution trace or evidence of engineering superiority.

269 seconds. This test demonstrates one successful recursive execution path. It predates the taxonomy integration and does not validate engineering performance or the specialist profiles.

6 PLANNED EVALUATION

The primary outcome will be independently scored engineering task correctness. The proposed task families cover CAD, CFD, structural mechanics, controls, optics, and digital electronics. Tasks must specify source artifacts, units, tolerances, compatibility constraints, and independent expected outcomes. Problem-family splits will separate related designs to reduce answer leakage. Answers and scorer artifacts must remain outside model-accessible memory.

The planned conditions are the same model without external resources, progressive skill disclosure, flat retrieval over the memory stores, recursive access to skills, recursive access to all stores, and the full taxonomy-scoped configuration. Models, prompts, and resource ceilings will be matched. Leave-one-store-out and taxonomy-disabled controls will test whether apparent improvements arise from additional evidence or from the scoping mechanism.

The initial planning target is twenty tasks per domain with three repeated runs per condition. A pilot will estimate variance and cost before freezing the final held-out design. Paired outcomes will

be summarized with confidence intervals obtained by resampling problem families, with repeated runs aggregated within task. We will report domain-specific correctness, citation support, abstention, constraint violations, latency, token usage, recursion depth, cost, and all failed attempts. No comparative results are available in this draft.

7 LIMITATIONS

The current corpus is uneven across domains and includes descriptive catalog records that do not establish executable access. Lexical retrieval may miss relevant synonyms. Editorial category errors can exclude useful tools, while shared records can retain irrelevant material. Source provenance does not establish source accuracy. Recursive calls can propagate incorrect assumptions or spend substantial computation on unproductive decomposition. The observed smoke test provides no estimate of robustness, statistical uncertainty, or engineering validity.

The system extends upstream procedural skills and an existing recursive runtime. Establishing novelty requires a completed comparison with adjacent engineering systems and a clear empirical distinction from simply combining retrieval with recursion. Claims about superior performance, unlimited context, autonomous design qualification, or broad domain competence are not supported by the present evidence.

AI USE STATEMENT

Generative AI assisted with the conceptual framework, implementation, literature discovery and synthesis, taxonomy integration, experimental protocol, and manuscript drafting. The synthetic recursion fixture was used for software validation rather than as a scientific dataset. Human review of the manuscript, citations, experimental design, and implementation remains pending. Before submission, the authors must verify the AI-assisted artifacts and update this statement to accurately describe completed review and their responsibility for the final work.

ETHICS STATEMENT

Engineering recommendations can fail when units, constraints, or source assumptions are wrong. Catalog records and generated recommendations require independent validation before consequential use. Redistributed artifacts must respect source licenses and exclude credentials or private catalog contents. This draft reports no human-subject study.

REPRODUCIBILITY STATEMENT

The code includes deterministic tests and a recorded synthetic recursive execution trace. The taxonomy appendix is generated from the same versioned definitions used by the runtime. The accompanying evaluation protocol specifies run metadata and artifact requirements. A fully anonymous reproducibility bundle and completed comparative experiment traces remain to be prepared before submission.

REFERENCES

- Timothy Kassis, Vinayak Agarwal, Yuhuan He, Darshil Patel, and Aubrey M. Brueckner. Scientific agent skills: A library of procedural knowledge for research agents, 2026.
- Charles Packer, Sarah Wooders, Kevin Lin, Vivian Fang, Shishir G. Patil, and Joseph E. Gonzalez. Memgpt: Towards llms as operating systems, 2023.
- Alex Zhang, Tim Kraska, and Omar Khattab. Recursive language models, 2025.

A COMPLETE ENGINEERING TAXONOMY

These editorial categories support browsing and scoped retrieval. Membership does not establish tool qualification. Counts refer to the pinned catalog snapshot; overlapping memberships are allowed.

A.1 COMPUTATIONAL GEOMETRY AND CAD KERNELS

Domain: `geometry`. Primary tool records: 71.

ID	Subcategory	Count
geometry-solid-modeling-kernels-and-cad-bindings	Solid Modeling Kernels and CAD Bindings	13
geometry-mesh-and-polygon-geometry-processing	Mesh and Polygon Geometry Processing	23
geometry-implicit-volumetric-and-generative-geometry	Implicit Volumetric and Generative Geometry	6
geometry-geometry-translation-and-scene-interchange	Geometry Translation and Scene Interchange	26
geometry-spatial-and-computational-geometry	Spatial and Computational Geometry	4
geometry-point-clouds-and-3d-reconstruction	Point Clouds and 3D Reconstruction	2
geometry-cad-quality-validation-and-healing	CAD Quality Validation and Healing	3
geometry-curve-and-surface-construction	Curve and Surface Construction	1
geometry-geometric-constraints-and-parametric-solvers	Geometric Constraints and Parametric Solvers	2
geometry-engineering-model-visualization-sdks	Engineering Model Visualization SDKs	1
geometry-mesh-geometry-editing-and-reconstruction	Mesh Geometry Editing and Reconstruction	1

A.2 COMPUTER-AIDED DESIGN

Domain: `cad`. Primary tool records: 65.

ID	Subcategory	Count
cad-mechanical	Mechanical and Product Engineering	25
cad-drafting	General Drafting and Technical Documentation	11
cad-architecture	Architectural and Building Design	7
cad-industrial	Industrial Design and Surface Modeling	12
cad-plant	Plant and Factory Layout Design	3
cad-marine	Naval Architecture and Ship Design	1
cad-aeronautical	Aircraft Geometry Design	1
cad-tooling	Mold, Die and Manufacturing Tooling Design	6
cad-computational	Computational and Programmatic Design	9
cad-preparation	Geometry Preparation for Simulation and Manufacturing	3
cad-general-3d	General 3D Geometric Modeling	4
cad-interopability	CAD File Conversion and Interoperability	1

A.3 STRUCTURAL ENGINEERING

Domain: `structural`. Primary tool records: 160.

ID	Subcategory	Count
structural-buildings	Building Structural Design	17
structural-bridges	Bridge and Transportation Structures	30
structural-fea	Structural Analysis and Finite Element Workflows	90
structural-nonlinear	Nonlinear Mechanics, Damage and Fracture	22
structural-dynamics	Structural Dynamics, Seismic and Impact Analysis	21
structural-components	Structural Connections and Component Design	6
structural-composites	Composite and Specialized Structural Materials	8
structural-fatigue	Structural Fatigue and Durability	4
structural-foundations	Structural Foundation Design	1
structural-materials	Structural Material Models and Constitutive Libraries	2
structural-piping	Piping and Equipment Structural Analysis	2
structural-tensile-membranes-cable-nets-and-form-finding	Tensile Membranes Cable Nets and Form Finding	11
structural-construction-stages-and-time-dependent-analysis	Construction Stages and Time-Dependent Analysis	1
structural-prestressed-and-post-tensioned-concrete	Prestressed and Post-Tensioned Concrete	1
structural-composite-modeling-and-results-automation	Composite Modeling and Results Automation	3
structural-composite-and-metallic-structural-sizing	Composite and Metallic Structural Sizing	1
structural-crash-impact-and-explicit-dynamics	Crash Impact and Explicit Dynamics	1

A.4 BUILDING INFORMATION MODELING

Domain: `bim`. Primary tool records: 57.

ID	Subcategory	Count
bim-bim-authoring-and-structural-detailing	BIM Authoring and Structural Detailing	14
bim-bim-coordination-and-model-checking	BIM Coordination and Model Checking	12
bim-bim-interoperability-and-developer-libraries	BIM Interoperability and Developer Libraries	7
bim-bim-collaboration-and-model-viewing	BIM Collaboration and Model Viewing	18
bim-construction-planning-and-4d-sequencing	Construction Planning and 4D Sequencing	1
bim-building-life-cycle-assessment-and-embodied-impacts	Building Life-Cycle Assessment and Embodied Impacts	3
bim-architectural-space-planning-and-design-feasibility	Architectural Space Planning and Design Feasibility	3
bim-mep-design-and-building-services-coordination	MEP Design and Building Services Coordination	2
bim-building-requirements-and-room-data-management	Building Requirements and Room Data Management	1
bim-facility-digital-twins-and-asset-information	Facility Digital Twins and Asset Information	1
bim-civil-infrastructure-conceptual-design	Civil Infrastructure Conceptual Design	1

A.5 GEOTECHNICAL ENGINEERING

Domain: `geotechnical`. Primary tool records: 70.

ID	Subcategory	Count
geotechnical-geotechnical-finite-element-and-continuum-analysis	Geotechnical Finite-Element and Continuum Analysis	21
geotechnical-slope-stability-and-groundwater-seepage	Slope Stability and Groundwater Seepage	24
geotechnical-foundations-retaining-structures-and-excavations	Foundations Retaining Structures and Excavations	23
geotechnical-rock-mechanics-tunneling-and-discontinuities	Rock Mechanics Tunneling and Discontinuities	15
geotechnical-geotechnical-investigation-and-ground-models	Geotechnical Investigation and Ground Models	1
geotechnical-seismic-site-response-and-liquefaction	Seismic Site Response and Liquefaction	2
geotechnical-discrete-element-and-granular-geomechanics	Discrete-Element and Granular Geomechanics	1
geotechnical-mine-geomechanics-and-material-flow	Mine Geomechanics and Material Flow	2
geotechnical-reinforced-soils-and-geosynthetic-structures	Reinforced Soils and Geosynthetic Structures	2

A.6 COMPUTATIONAL FLUID DYNAMICS

Domain: `cf-d`. Primary tool records: 131.

ID	Subcategory	Count
cf-d-aerodynamics	Aerodynamics and External Flow	10
cf-d-multiphase	Multiphase and Free-Surface Flow	19
cf-d-combustion	Combustion and Reacting Flow	20
cf-d-compressible	Compressible Flow and Gas Dynamics	13
cf-d-particles	Particle and Kinetic Flow Methods	29
cf-d-thermal	Thermal Fluids and Conjugate Heat Transfer	20
cf-d-turbomachinery	Turbomachinery and Rotating Flow	5
cf-d-general	CFD Solvers and Flow Modeling Frameworks	49
cf-d-marine	Marine and Coastal Hydrodynamics	9
cf-d-fsi	Fluid-Structure Interaction	1
cf-d-fire	Fire Simulation and Evacuation Workflows	5
cf-d-electrohydrodynamics-and-coupled-field-flows	Electrohydrodynamics and Coupled Field Flows	1

A.7 RADIO FREQUENCY AND ELECTROMAGNETIC ENGINEERING

Domain: `rf-em`. Primary tool records: 96.

ID	Subcategory	Count
rf-em-antennas	Antenna Design and Electromagnetic Scattering	14
rf-em-microwave	Microwave, RF and High-Frequency Circuit Design	20
rf-em-compatibility	Electromagnetic Compatibility and Interference	6
rf-em-low-frequency	Low-Frequency Fields and Electromechanics	16
rf-em-fullwave	Full-Wave Electromagnetic Simulation	22
rf-em-interconnect	Signal Integrity and Electronic Interconnects	16

ID	Subcategory	Count
rf-em-quantum	Quantum Transport and Electronic Device Modeling	15
rf-em-superconducting	Superconducting and Josephson Circuit Modeling	7
rf-em-layout	Microelectronic and Photonic Layout Libraries	3
rf-em-photonics	Optical and Photonic Electromagnetic Modeling	2
rf-em-propagation	Wireless Propagation and Network Planning	1

A.8 OPTICAL ENGINEERING AND PHOTONICS

Domain: `optics`. Primary tool records: 167.

ID	Subcategory	Count
optics-imaging	Optical System Design and Imaging	33
optics-illumination	Illumination and Nonimaging Optics	17
optics-wave	Wave Optics and Electromagnetic Propagation	52
optics-integrated	Integrated Photonics and Photonic Circuits	13
optics-laser	Lasers, Ultrafast and Nonlinear Optics	35
optics-fiber	Fiber Optics and Optical Communications	17
optics-coatings	Optical Coatings and Thin Films	2
optics-characterization	Optical Measurement and Characterization	5
optics-devices	Optoelectronic and Semiconductor Device Modeling	1
optics-inverse	Optical Optimization and Inverse Design	10
optics-adaptive	Adaptive Optics and Wavefront Control	10
optics-optomechanical-and-thermal-performance-integration	Optomechanical and Thermal Performance Integration	5

A.9 INTEGRATED CIRCUIT DESIGN AND SEMICONDUCTOR DEVICES

Domain: `eda-digital`. Primary tool records: 55.

ID	Subcategory	Count
eda-digital-logic-synthesis-and-physical-implementation	Logic Synthesis and Physical Implementation	13
eda-digital-timing-power-and-signoff-analysis	Timing Power and Signoff Analysis	8
eda-digital-digital-simulation-and-verification	Digital Simulation and Verification	27
eda-digital-test-design-and-silicon-debug	Test Design and Silicon Debug	6
eda-digital-custom-layout-and-physical-verification	Custom Layout and Physical Verification	3
eda-digital-eda-flow-orchestration-and-build-infrastructure	EDA Flow Orchestration and Build Infrastructure	2
eda-digital-microfabrication-and-mask-layout	Microfabrication and Mask Layout	1
eda-digital-semiconductor-device-simulation	Semiconductor Device Simulation	1

A.10 FPGA DESIGN AND VERIFICATION

Domain: `fpga`. Primary tool records: 34.

ID	Subcategory	Count
fpga-fpga-synthesis-and-place-and-route	FPGA Synthesis and Place-and-Route	19
fpga-high-level-synthesis-and-embedded-software	High-Level Synthesis and Embedded Software	4
fpga-fpga-simulation-verification-and-debug	FPGA Simulation Verification and Debug	12
fpga-fpga-programming-and-hardware-integration	FPGA Programming and Hardware Integration	12

A.11 ELECTRONIC CIRCUIT SIMULATION

Domain: `spice`. Primary tool records: 39.

ID	Subcategory	Count
spice-spice-and-analog-circuit-simulation	SPICE and Analog Circuit Simulation	31
spice-power-electronics-and-switching-systems	Power Electronics and Switching Systems	3
spice-mixed-signal-and-behavioral-modeling	Mixed-Signal and Behavioral Modeling	4
spice-circuit-design-automation-and-schematic-interfaces	Circuit Design Automation and Schematic Interfaces	7
spice-rf-and-microwave-circuit-simulation	RF and Microwave Circuit Simulation	4

A.12 MANUFACTURING ENGINEERING

Domain: `manufacturing`. Primary tool records: 203.

ID	Subcategory	Count
manufacturing-machining	CNC Machining and CAM	62
manufacturing-additive	Additive Manufacturing	23
manufacturing-forming	Metal Forming and Sheet Metal Processing	22
manufacturing-molding	Casting and Polymer Processing	22
manufacturing-joining	Welding and Joining Processes	6
manufacturing-thermal	Heat Treatment and Surface Processing	14
manufacturing-planning	Manufacturing Process Planning and Operations	4
manufacturing-composites	Composite Manufacturing	16
manufacturing-simulation	Manufacturing Process Simulation and Optimization	13
manufacturing-preparation	Manufacturing Geometry and Job Preparation	2
manufacturing-response	Materials Response and Microstructure for Manufacturing	3
manufacturing-fluid	Process Fluid Simulation	1
manufacturing-polymer-processing-and-extrusion-simulation	Polymer Processing and Extrusion Simulation	1
manufacturing-electronics-encapsulation-and-packaging-processes	Electronics Encapsulation and Packaging Processes	2
manufacturing-composite-forming-draping-and-layup-preparation	Composite Forming Draping and Layup Preparation	4
manufacturing-metal-forming-and-process-simulation	Metal Forming and Process Simulation	8
manufacturing-heat-treatment-and-microstructure-evolution	Heat Treatment and Microstructure Evolution	2

ID	Subcategory	Count
manufacturing-filament-winding-and-composite-toolpaths	Filament Winding and Composite Toolpaths	2
manufacturing-manufacturing-process-planning	Manufacturing Process Planning	1
manufacturing-textile-architecture-weaving-and-fabric-simulation	Textile Architecture Weaving and Fabric Simulation	10
manufacturing-membrane-cutting-patterns-and-fabrication	Membrane Cutting Patterns and Fabrication	2
manufacturing-composite-infusion-and-permeability	Composite Infusion and Permeability	2
manufacturing-casting-and-solidification	Casting and Solidification	3
manufacturing-automated-fiber-placement-and-tape-laying	Automated Fiber Placement and Tape Laying	6
manufacturing-composite-cure-and-process-induced-distortion	Composite Cure and Process-Induced Distortion	1
manufacturing-composite-layup-design-and-manufacturing-preparation	Composite Layup Design and Manufacturing Preparation	2
manufacturing-composite-material-tracking-and-production-management	Composite Material Tracking and Production Management	1
manufacturing-sheet-cutting-nesting-and-cnc-programming	Sheet Cutting Nesting and CNC Programming	1

A.13 MATERIALS SCIENCE AND ENGINEERING

Domain: `materials`. Primary tool records: 96.

ID	Subcategory	Count
materials-thermodynamics	Materials Thermodynamics and Phase Equilibria	7
materials-microstructure	Microstructure and Phase-Field Modeling	27
materials-constitutive	Constitutive Modeling and Material Response	10
materials-processing	Materials Processing and Heat Treatment	5
materials-crystallography	Crystallography, Diffraction and Scattering	27
materials-magnetism	Micromagnetics and Magnetic Materials	13
materials-discovery	Materials Discovery and Informatics	3
materials-composites	Composite and Multiscale Materials	2
materials-kinetics	Surface Chemistry and Kinetic Modeling	4
materials-data	Materials Data, Structure Preparation and Characterization	4
materials-potentials	Interatomic Potentials and Atomistic Material Models	1
materials-composite-micromechanics-and-homogenization	Composite Micromechanics and Homogenization	2
materials-thermal-protection-and-ablative-material-response	Thermal Protection and Ablative Material Response	1

A.14 ROBOTICS ENGINEERING

Domain: `robotics`. Primary tool records: 68.

ID	Subcategory	Count
robotics-programming	Industrial Robot Programming and Workcells	27

ID	Subcategory	Count
robotics-simulation	Robot Simulation and Virtual Environments	21
robotics-dynamics	Robot Kinematics, Dynamics and Contact Physics	16
robotics-planning	Motion Planning and Trajectory Optimization	13
robotics-learning	Robot Learning and Synthetic Data	2
robotics-middleware	Robotics Middleware and Application Frameworks	4
robotics-commissioning	Manufacturing Simulation and Virtual Commissioning	7
robotics-education	Robotics Education and Algorithm Prototyping	1
robotics-small-farm	Small-Farm Automation	1
robotics-fleet-coordination-and-facility-integration	Fleet Coordination and Facility Integration	1

A.15 ELECTRICAL POWER SYSTEMS ENGINEERING

Domain: `power`. Primary tool records: 51.

ID	Subcategory	Count
power-power-flow-distribution-and-network-planning	Power Flow Distribution and Network Planning	24
power-electromagnetic-transients-and-real-time-simulation	Electromagnetic Transients and Real-Time Simulation	9
power-protection-fault-and-electrical-safety-analysis	Protection Fault and Electrical Safety Analysis	12
power-grid-dynamics-stability-and-frequency-control	Grid Dynamics Stability and Frequency Control	7
power-electric-machine-and-motor-design	Electric Machine and Motor Design	1
power-energy-markets-resource-adequacy-and-capacity-expansion	Energy Markets Resource Adequacy and Capacity Expansion	13
power-railway-traction-power-and-electrification	Railway Traction Power and Electrification	2

A.16 SCIENTIFIC MACHINE LEARNING

Domain: `sciml`. Primary tool records: 49.

ID	Subcategory	Count
sciml-physics-informed-neural-networks	Physics-Informed Neural Networks	6
sciml-neural-operators-and-learned-surrogates	Neural Operators and Learned Surrogates	10
sciml-differentiable-simulation-and-scientific-solvers	Differentiable Simulation and Scientific Solvers	26
sciml-machine-learning-frameworks-and-regression	Machine Learning Frameworks and Regression	13
sciml-symbolic-regression-and-scientific-programming	Symbolic Regression and Scientific Programming	2

A.17 LABORATORY AUTOMATION AND INSTRUMENTATION

Domain: `lab`. Primary tool records: 69.

ID	Subcategory	Count
lab-control	Instrument Control and Data Acquisition	52
lab-automation	Experiment Orchestration and Laboratory Automation	14
lab-processing	Experimental Data Processing and Analysis	3
lab-integration	Instrument Communication and Integration Libraries	12

A.18 BUILDING ENERGY AND HVAC ENGINEERING

Domain: `building-energy`. Primary tool records: 56.

ID	Subcategory	Count
building-energy-whole-building-energy-simulation	Whole-Building Energy Simulation	15
building-energy-district-and-urban-energy-modeling	District and Urban Energy Modeling	9
building-energy-hvac-systems-and-building-controls	HVAC Systems and Building Controls	27
building-energy-envelope-moisture-and-passive-design	Envelope Moisture and Passive Design	8
building-energy-daylighting-and-early-design-assessment	Daylighting and Early Design Assessment	8
building-energy-energy-code-compliance-and-ratings	Energy Code Compliance and Ratings	3
building-energy-building-airflow-and-indoor-air-quality	Building Airflow and Indoor Air Quality	1

A.19 WATER RESOURCES AND ENVIRONMENTAL MODELING

Domain: `water`. Primary tool records: 112.

ID	Subcategory	Count
water-hydrology	Watershed Hydrology and Rainfall-Runoff Modeling	29
water-rivers	River Hydraulics and Flood Modeling	34
water-drainage	Urban Drainage and Sewer Systems	7
water-distribution	Water Distribution and Pipe Networks	10
water-quality	Water Quality and Aquatic Ecosystems	11
water-coastal	Coastal, Estuarine and Ocean Hydrodynamics	34
water-groundwater	Groundwater and Integrated Water Resources	19
water-sediment	Sediment Transport and Morphodynamics	5
water-hydraulic-model-preparation-and-visualization	Hydraulic Model Preparation and Visualization	1
water-hydrological-model-preparation-and-gis-integration	Hydrological Model Preparation and GIS Integration	1
water-hydrologic-frequency-and-dam-risk-analysis	Hydrologic Frequency and Dam-Risk Analysis	2

A.20 ELECTROCHEMICAL ENGINEERING

Domain: `electrochemistry`. Primary tool records: 59.

ID	Subcategory	Count
electrochemistry-battery-cell-and-pack-modeling	Battery Cell and Pack Modeling	29

ID	Subcategory	Count
electrochemistry-fuel-cells-electrolyzers-and-hydrogen-systems	Fuel Cells Electrolyzers and Hydrogen Systems	15
electrochemistry-electrochemical-kinetics-and-transport	Electrochemical Kinetics and Transport	6
electrochemistry-electrochemical-instrument-control-and-experiment-analysis	Electrochemical Instrument Control and Experiment Analysis	1
electrochemistry-impedance-spectroscopy-and-equivalent-circuit-analysis	Impedance Spectroscopy and Equivalent-Circuit Analysis	3
electrochemistry-battery-manufacturing-and-cost-analysis	Battery Manufacturing and Cost Analysis	1
electrochemistry-hydrogen-production-delivery-and-refueling-economics	Hydrogen Production Delivery and Refueling Economics	6
electrochemistry-battery-aging-and-lifetime-prediction	Battery Aging and Lifetime Prediction	1

A.21 CHEMICAL AND PROCESS ENGINEERING

Domain: `chemical`. Primary tool records: 69.

ID	Subcategory	Count
chemical-process-flowsheeting-and-plant-simulation	Process Flowsheeting and Plant Simulation	34
chemical-reaction-kinetics-and-reactor-modeling	Reaction Kinetics and Reactor Modeling	15
chemical-thermophysical-properties-and-transport	Thermophysical Properties and Transport	16
chemical-separation-processes-and-unit-operations	Separation Processes and Unit Operations	6
chemical-batch-and-particulate-process-engineering	Batch and Particulate Process Engineering	7
chemical-chemical-structure-representation-and-cheminformatics	Chemical Structure Representation and Cheminformatics	5
chemical-mixing-and-agitated-process-equipment	Mixing and Agitated Process Equipment	1
chemical-chemical-kinetics-and-mechanism-reduction	Chemical Kinetics and Mechanism Reduction	1
chemical-chemical-equilibrium-and-thermochemistry	Chemical Equilibrium and Thermochemistry	1

A.22 POROUS MEDIA AND SUBSURFACE FLOW

Domain: `porous-media`. Primary tool records: 76.

ID	Subcategory	Count
porous-media-reservoir-and-multiphase-flow-simulation	Reservoir and Multiphase Flow Simulation	22
porous-media-reactive-transport-and-geochemistry	Reactive Transport and Geochemistry	12
porous-media-groundwater-solute-and-unsaturated-transport	Groundwater Solute and Unsaturated Transport	9
porous-media-pore-scale-and-porous-media-modeling	Pore-Scale and Porous-Media Modeling	1

ID	Subcategory	Count
porous-media-fractured-media-flow-and-geomechanics	Fractured-Media Flow and Geomechanics	1
porous-media-continuum-porous-media-flow-and-transport	Continuum Porous-Media Flow and Transport	3
porous-media-pore-scale-flow-and-multiphase-transport	Pore-Scale Flow and Multiphase Transport	2
porous-media-reservoir-history-matching-uncertainty-and-optimization	Reservoir History Matching Uncertainty and Optimization	1
porous-media-reservoir-wellbore-and-production-system-coupling	Reservoir Wellbore and Production-System Coupling	5
porous-media-subsurface-engineering-platforms-and-data-integration	Subsurface Engineering Platforms and Data Integration	1
porous-media-drilling-well-control-and-wellbore-hydraulics	Drilling Well Control and Wellbore Hydraulics	1
porous-media-geological-carbon-storage-assessment	Geological Carbon Storage Assessment	1
porous-media-hydraulic-fracturing-and-well-stimulation	Hydraulic Fracturing and Well Stimulation	5
porous-media-well-testing-production-diagnostics-and-surveillance	Well Testing Production Diagnostics and Surveillance	5
porous-media-well-design-completion-and-production-hydraulics	Well Design Completion and Production Hydraulics	3
porous-media-well-design-and-integrity	Well Design and Integrity	4
porous-media-reservoir-model-automation-and-interfaces	Reservoir Model Automation and Interfaces	1
porous-media-well-lifecycle-data-and-operations	Well Lifecycle Data and Operations	2
porous-media-reservoir-model-preparation-and-visualization	Reservoir Model Preparation and Visualization	2

A.23 MULTIPHYSICS MODELING AND SIMULATION

Domain: `multiphysics`. Primary tool records: 51.

ID	Subcategory	Count
multiphysics-pde-and-finite-element-frameworks	PDE and Finite-Element Frameworks	28
multiphysics-multiphysics-simulation-suites	Multiphysics Simulation Suites	8
multiphysics-coupling-and-co-simulation-infrastructure	Coupling and Co-Simulation Infrastructure	3
multiphysics-multibody-and-mechanism-simulation	Multibody and Mechanism Simulation	6
multiphysics-thermal-and-electromagnetic-systems	Thermal and Electromagnetic Systems	5
multiphysics-flexible-rods-cables-and-slender-body-mechanics	Flexible Rods Cables and Slender-Body Mechanics	1
multiphysics-discrete-element-and-granular-mechanics	Discrete-Element and Granular Mechanics	2

A.24 ACOUSTICS AND VIBRATION ENGINEERING

Domain: `acoustics`. Primary tool records: 57.

ID	Subcategory	Count
acoustics-room-and-architectural-acoustics	Room and Architectural Acoustics	17
acoustics-environmental-noise-and-sound-propagation	Environmental Noise and Sound Propagation	4
acoustics-vibroacoustics-and-structural-sound	Vibroacoustics and Structural Sound	6
acoustics-acoustic-wave-solvers-and-signal-analysis	Acoustic Wave Solvers and Signal Analysis	10
acoustics-absorbers-and-acoustic-materials	Absorbers and Acoustic Materials	3
acoustics-electroacoustic-systems-and-loudspeakers	Electroacoustic Systems and Loudspeakers	2
acoustics-underwater-acoustics-and-sonar	Underwater Acoustics and Sonar	10
acoustics-acoustic-measurement-and-source-localization	Acoustic Measurement and Source Localization	6
acoustics-thermoacoustics-and-combustion-instability	Thermoacoustics and Combustion Instability	1
acoustics-ultrasound-and-piezoelectric-transducers	Ultrasound and Piezoelectric Transducers	1

A.25 RENEWABLE ENERGY ENGINEERING

Domain: `renewables`. Primary tool records: 50.

ID	Subcategory	Count
renewables-wind-turbine-aerodynamics-and-structural-dynamics	Wind Turbine Aerodynamics and Structural Dynamics	6
renewables-wind-resource-assessment-and-farm-layout	Wind Resource Assessment and Farm Layout	10
renewables-solar-photovoltaic-design-and-yield	Solar Photovoltaic Design and Yield	14
renewables-marine-renewable-energy-systems	Marine Renewable Energy Systems	8
renewables-energy-system-planning-and-technoeconomic-analysis	Energy System Planning and Technoeconomic Analysis	10
renewables-energy-storage-and-distributed-energy-systems	Energy Storage and Distributed Energy Systems	7

A.26 AERONAUTICAL ENGINEERING

Domain: `aerospace`. Primary tool records: 122.

ID	Subcategory	Count
aerospace-aircraft-conceptual-design-and-sizing	Aircraft Conceptual Design and Sizing	20
aerospace-aerodynamics-and-aeroelasticity	Aerodynamics and Aeroelasticity	25
aerospace-flight-dynamics-and-simulation	Flight Dynamics and Simulation	12

ID	Subcategory	Count
aerospace-propulsion-and-engine-performance	Propulsion and Engine Performance	50
aerospace-rotorcraft-and-rotor-aerodynamics	Rotorcraft and Rotor Aerodynamics	9
aerospace-aeroacoustics-and-propulsion-noise	Aeroacoustics and Propulsion Noise	10
aerospace-flight-control-and-handling-qualities	Flight Control and Handling Qualities	13
aerospace-rocket-design-and-flight-simulation	Rocket Design and Flight Simulation	4
aerospace-aircraft-performance-and-mission-analysis	Aircraft Performance and Mission Analysis	3
aerospace-open-aircraft-designs	Open Aircraft Designs and Build Resources	1

A.27 SPACE SYSTEMS AND ASTRONAUTICAL ENGINEERING

Domain: `space`. Primary tool records: 97.

ID	Subcategory	Count
space-orbital-mechanics-and-trajectory-design	Orbital Mechanics and Trajectory Design	42
space-spacecraft-dynamics-guidance-and-control	Spacecraft Dynamics Guidance and Control	9
space-spacecraft-thermal-and-environmental-analysis	Spacecraft Thermal and Environmental Analysis	28
space-mission-systems-communications-and-coverage	Mission Systems Communications and Coverage	9
space-mission-operations-and-flight-software-testing	Mission Operations and Flight Software Testing	8
space-plume-interaction-and-surface-operations	Plume Interaction and Surface Operations	1
space-planetary-surface-operations-and-resource-utilization	Planetary Surface Operations and Resource Utilization	2
space-spacecraft-power-and-energy-systems	Spacecraft Power and Energy Systems	3
space-exoplanet-mission-design-and-yield-analysis	Exoplanet Mission Design and Yield Analysis	1

A.28 NUCLEAR SCIENCE AND ENGINEERING

Domain: `nuclear`. Primary tool records: 117.

ID	Subcategory	Count
nuclear-transport	Radiation Transport and Shielding	29
nuclear-reactor	Reactor Physics and Core Analysis	44
nuclear-thermal	Nuclear Thermal Hydraulics and Safety Analysis	19
nuclear-fuel	Nuclear Fuel Performance and Fuel Cycles	30
nuclear-fusion	Fusion Engineering and Neutronics	2
nuclear-environment	Environmental Radiological Assessment and Dosimetry	7
nuclear-risk	Nuclear Reliability and Probabilistic Risk Assessment	1
nuclear-integration	Nuclear Analysis Integration and Workflow	8
nuclear-nuclear-data-processing-and-evaluation	Nuclear Data Processing and Evaluation	6

ID	Subcategory	Count
nuclear-transport-benchmarking-and-nuclear-data-validation	Transport Benchmarking and Nuclear-Data Validation	1
nuclear-neutron-instrument-and-scattering-simulation	Neutron Instrument and Scattering Simulation	2
nuclear-radiation-transport-geometry-and-model-preparation	Radiation Transport Geometry and Model Preparation	1

A.29 NUMERICAL METHODS AND SCIENTIFIC COMPUTING

Domain: `numerical`. Primary tool records: 90.

ID	Subcategory	Count
numerical-linear	Linear Algebra and Matrix Computation	22
numerical-differential	Differential Equations and Numerical Solvers	8
numerical-symbolic	Symbolic and Computer Algebra	11
numerical-arrays	Numerical Arrays and Scientific Computing	14
numerical-transforms	Transforms, Approximation and Numerical Analysis	4
numerical-acceleration	Parallel and Accelerated Numerical Computing	16
numerical-precision	Arbitrary-Precision and Interval Arithmetic	7
numerical-optimization	Numerical Optimization and Mathematical Programming	7
numerical-differentiation	Automatic Differentiation	7
numerical-compilation	Scientific Compilation and Code Acceleration	2
numerical-uncertainty	Uncertainty Quantification and Propagation	2
numerical-storage	Scientific Data Storage and Interchange	1
numerical-units	Physical Units and Dimensional Computation	1
numerical-formal-mathematics-theorem-proving-and-smt	Formal Mathematics Theorem Proving and SMT	7

A.30 SCIENTIFIC DATA ANALYSIS AND STATISTICS

Domain: `data-analysis`. Primary tool records: 71.

ID	Subcategory	Count
data-analysis-tabular-and-multidimensional-data-processing	Tabular and Multidimensional Data Processing	23
data-analysis-scientific-storage-and-data-interchange	Scientific Storage and Data Interchange	29
data-analysis-statistics-and-bayesian-inference	Statistics and Bayesian Inference	22
data-analysis-machine-learning-and-model-interpretation	Machine Learning and Model Interpretation	3
data-analysis-distributed-and-large-scale-data-computing	Distributed and Large-Scale Data Computing	2
data-analysis-time-series-analysis-and-forecasting	Time-Series Analysis and Forecasting	4
data-analysis-time-series-and-forecasting	Time Series and Forecasting	1

A.31 MATHEMATICAL OPTIMIZATION

Domain: `optimization`. Primary tool records: 91.

ID	Subcategory	Count
optimization-mathematical-programming-and-constraint-solvers	Mathematical Programming and Constraint Solvers	27
optimization-engineering-design-and-multidisciplinary-optimization	Engineering Design and Multidisciplinary Optimization	24
optimization-global-evolutionary-and-black-box-optimization	Global Evolutionary and Black-Box Optimization	16
optimization-uncertainty-sensitivity-and-experimental-design	Uncertainty Sensitivity and Experimental Design	16
optimization-surrogate-modeling-and-parameter-fitting	Surrogate Modeling and Parameter Fitting	12
optimization-optimal-control-and-differentiable-optimization	Optimal Control and Differentiable Optimization	7
optimization-design-study-and-optimization-workflow-orchestration	Design Study and Optimization Workflow Orchestration	2

A.32 MESH GENERATION AND PROCESSING

Domain: meshing. Primary tool records: 48.

ID	Subcategory	Count
meshing-finite-element-and-volume-mesh-generation	Finite-Element and Volume Mesh Generation	37
meshing-mesh-adaptation-optimization-and-repair	Mesh Adaptation Optimization and Repair	11
meshing-mesh-interchange-and-preprocessing	Mesh Interchange and Preprocessing	2
meshing-specialized-cfd-and-boundary-layer-meshing	Specialized CFD and Boundary-Layer Meshing	6
meshing-thermal-model-geometry-preparation	Thermal Model Geometry Preparation	1

A.33 PRINTED CIRCUIT BOARD DESIGN

Domain: eda-pcb. Primary tool records: 66.

ID	Subcategory	Count
eda-pcb-pcb-schematic-and-layout-design	PCB Schematic and Layout Design	26
eda-pcb-signal-power-and-electromagnetic-integrity	Signal Power and Electromagnetic Integrity	13
eda-pcb-advanced-packaging-and-heterogeneous-integration	Advanced Packaging and Heterogeneous Integration	8
eda-pcb-pcb-manufacturing-and-assembly-verification	PCB Manufacturing and Assembly Verification	17
eda-pcb-electronics-reliability-and-thermal-analysis	Electronics Reliability and Thermal Analysis	2
eda-pcb-pcb-stackup-design-and-engineering-calculators	PCB Stackup Design and Engineering Calculators	5
eda-pcb-electronics-thermal-management	Electronics Thermal Management	5
eda-pcb-behavioral-i-o-and-ibis-model-development	Behavioral I/O and IBIS Model Development	1
eda-pcb-package-thermomechanical-reliability	Package Thermomechanical Reliability	2

ID	Subcategory	Count
eda-pcb-electronic-design-governance-and-collaboration	Electronic Design Governance and Collaboration	1

A.34 CONTROL SYSTEMS AND DYNAMIC SIMULATION

Domain: `controls`. Primary tool records: 140.

ID	Subcategory	Count
controls-physical-system-and-modelica-simulation	Physical System and Modelica Simulation	25
controls-control-design-and-optimal-control	Control Design and Optimal Control	17
controls-co-simulation-and-model-exchange	Co-Simulation and Model Exchange	11
controls-real-time-simulation-and-hardware-in-the-loop	Real-Time Simulation and Hardware-in-the-Loop	20
controls-system-identification-and-state-estimation	System Identification and State Estimation	6
controls-state-machines-and-discrete-event-control	State Machines and Discrete-Event Control	2
controls-differential-equations-and-dynamic-system-solvers	Differential Equations and Dynamic System Solvers	4
controls-mechanical-circuit-and-educational-system-simulation	Mechanical Circuit and Educational System Simulation	1
controls-embedded-control-software-and-code-generation	Embedded Control Software and Code Generation	4
controls-vehicle-and-sensor-simulation	Vehicle and Sensor Simulation	37
controls-rail-vehicle-and-train-dynamics	Rail Vehicle and Train Dynamics	1
controls-multibody-dynamics-and-mechanism-simulation	Multibody Dynamics and Mechanism Simulation	23
controls-industrial-monitoring-and-supervisory-control	Industrial Monitoring and Supervisory Control	3
controls-plc-programming-and-motion-control	PLC Programming and Motion Control	4

A.35 STATISTICAL MECHANICS

Domain: `statistical-mechanics`. Primary tool records: 44.

ID	Subcategory	Count
statistical-mechanics-molecular-and-particle-dynamics	Molecular and Particle Dynamics	10
statistical-mechanics-monte-carlo-and-ensemble-sampling	Monte Carlo and Ensemble Sampling	17
statistical-mechanics-enhanced-sampling-and-free-energy	Enhanced Sampling and Free Energy	11
statistical-mechanics-trajectory-analysis-and-statistical-estimation	Trajectory Analysis and Statistical Estimation	2
statistical-mechanics-mesoscopic-and-reaction-diffusion-simulation	Mesoscopic and Reaction-Diffusion Simulation	2
statistical-mechanics-thermodynamics-and-molecular-property-prediction	Thermodynamics and Molecular Property Prediction	1

ID	Subcategory	Count
statistical-mechanics-magnetic-structures-and-spin-dynamics	Magnetic Structures and Spin Dynamics	4

A.36 ATOMISTIC MODELING AND QUANTUM CHEMISTRY

Domain: `atomistic`. Primary tool records: 85.

ID	Subcategory	Count
atomistic-electronic	Electronic Structure and Density Functional Theory	42
atomistic-quantum-chemistry	Molecular Quantum Chemistry	20
atomistic-molecular-dynamics	Molecular Dynamics and Atomistic Simulation	15
atomistic-potentials	Interatomic Potentials and Machine Learning	9
atomistic-workflows	Atomistic Workflows and Structure Analysis	21
atomistic-transport	Electronic Transport and Mesoscopic Modeling	4
atomistic-excited	Excited-State and Nonadiabatic Dynamics	1
atomistic-vibrations	Lattice Dynamics and Phonons	1
atomistic-integrated	Integrated Molecular and Materials Modeling	2
atomistic-molecular-force-fields-and-parameterization	Molecular Force Fields and Parameterization	1

A.37 THERMODYNAMICS AND THERMOPHYSICAL PROPERTIES

Domain: `thermophysical`. Primary tool records: 99.

ID	Subcategory	Count
thermophysical-fluid-properties-and-equations-of-state	Fluid Properties and Equations of State	18
thermophysical-thermal-hydraulic-networks-and-transients	Thermal Hydraulic Networks and Transients	39
thermophysical-heat-exchangers-and-thermal-equipment	Heat Exchangers and Thermal Equipment	7
thermophysical-thermodynamic-cycles-and-energy-systems	Thermodynamic Cycles and Energy Systems	10
thermophysical-cryogenic-and-vacuum-systems	Cryogenic and Vacuum Systems	20
thermophysical-psychrometrics-and-moist-air-properties	Psychrometrics and Moist-Air Properties	1
thermophysical-heat-transfer-simulation	Heat Transfer Simulation	4
thermophysical-thermal-model-automation-and-interfaces	Thermal Model Automation and Interfaces	1
thermophysical-ground-source-heat-and-geothermal-borefields	Ground-Source Heat and Geothermal Borefields	2
thermophysical-cryogenic-systems-and-superconducting-magnet-thermal-analysis	Cryogenic Systems and Superconducting Magnet Thermal Analysis	4
thermophysical-heat-exchanger-design-and-rating	Heat Exchanger Design and Rating	1

A.38 SCIENTIFIC VISUALIZATION

Domain: `visualization`. Primary tool records: 58.

ID	Subcategory	Count
visualization-scientific-3d-visualization-and-postprocessing	Scientific 3D Visualization and Postprocessing	27
visualization-plotting-charts-and-interactive-dashboards	Plotting Charts and Interactive Dashboards	21
visualization-visualization-libraries-and-web-graphics	Visualization Libraries and Web Graphics	5
visualization-specialized-scientific-data-viewers	Specialized Scientific Data Viewers	13

A.39 RENDERING AND COMPUTER GRAPHICS

Domain: `rendering`. Primary tool records: 93.

ID	Subcategory	Count
rendering-production	Production and Photorealistic Rendering	28
rendering-architecture	Architectural Visualization	8
rendering-product	Product and Industrial Visualization	12
rendering-realtime	Real-Time Rendering and Interactive Graphics	20
rendering-scientific	Scientific, Spectral and Differentiable Rendering	19
rendering-infrastructure	Rendering Libraries and Scene Infrastructure	22
rendering-content	3D Scene Creation and Animation	3
rendering-neural	Neural Rendering and Synthetic Scene Generation	8

A.40 OPERATIONS RESEARCH AND DISCRETE-EVENT SIMULATION

Domain: `operations`. Primary tool records: 46.

ID	Subcategory	Count
operations-discrete-event-and-queueing-simulation	Discrete-Event and Queueing Simulation	24
operations-factory-production-and-material-flow	Factory Production and Material Flow	7
operations-supply-chain-logistics-and-scheduling	Supply Chain Logistics and Scheduling	13
operations-agent-based-and-hybrid-operations-modeling	Agent-Based and Hybrid Operations Modeling	2
operations-hosted-simulation-and-experiment-execution	Hosted Simulation and Experiment Execution	1
operations-factory-layout-and-equipment-planning	Factory Layout and Equipment Planning	2
operations-workstation-ergonomics-and-human-factors	Workstation Ergonomics and Human Factors	2
operations-airport-capacity-and-air-traffic-simulation	Airport Capacity and Air Traffic Simulation	1
operations-engineering-project-portfolio-and-cost-management	Engineering Project Portfolio and Cost Management	2

A.41 ASTRONOMY AND ASTROPHYSICS

Domain: `astronomy`. Primary tool records: 120.

ID	Subcategory	Count
astronomy-astronomical-images-and-photometry	Astronomical Images and Photometry	11
astronomy-spectroscopy-and-radio-astronomy	Spectroscopy and Radio Astronomy	29
astronomy-observatory-control-and-observation-planning	Observatory Control and Observation Planning	17
astronomy-astronomical-catalogs-and-data-access	Astronomical Catalogs and Data Access	39
astronomy-cosmology-and-astrophysical-simulation	Cosmology and Astrophysical Simulation	17
astronomy-astronomical-data-reduction-and-statistical-modeling	Astronomical Data Reduction and Statistical Modeling	11
astronomy-astronomical-coordinates-time-and-numerical-utilities	Astronomical Coordinates Time and Numerical Utilities	1
astronomy-astrochemistry-and-radiative-transfer	Astrochemistry and Radiative Transfer	6
astronomy-telescope-instrument-and-observation-simulation	Telescope Instrument and Observation Simulation	3
astronomy-interstellar-chemistry-and-radiative-transfer	Interstellar Chemistry and Radiative Transfer	1

A.42 EXPERIMENTAL DATA ANALYSIS AND CHARACTERIZATION

Domain: experimental-analysis. Primary tool records: 67.

ID	Subcategory	Count
experimental-analysis-diffraction-crystallography-and-scattering	Diffraction Crystallography and Scattering	23
experimental-analysis-spectroscopy-and-spectral-analysis	Spectroscopy and Spectral Analysis	18
experimental-analysis-microscopy-and-hyperspectral-analysis	Microscopy and Hyperspectral Analysis	10
experimental-analysis-modal-vibration-and-acoustic-testing	Modal Vibration and Acoustic Testing	3
experimental-analysis-experimental-data-reduction-and-fitting	Experimental Data Reduction and Fitting	3
experimental-analysis-optical-flow-measurement-and-particle-velocimetry	Optical Flow Measurement and Particle Velocimetry	7
experimental-analysis-tomographic-reconstruction-and-microscopy-processing	Tomographic Reconstruction and Microscopy Processing	6
experimental-analysis-mass-spectrometry-and-chromatography	Mass Spectrometry and Chromatography	5
experimental-analysis-test-data-management-analysis-and-reporting	Test Data Management Analysis and Reporting	1

A.43 SCIENTIFIC IMAGING AND IMAGE PROCESSING

Domain: imaging. Primary tool records: 48.

ID	Subcategory	Count
imaging-image-processing-and-computer-vision	Image Processing and Computer Vision	20

ID	Subcategory	Count
imaging-tomography-and-inverse-reconstruction	Tomography and Inverse Reconstruction	22
imaging-scientific-image-formats-and-acquisition	Scientific Image Formats and Acquisition	9
imaging-3d-photogrammetry-and-image-based-metrology	3D Photogrammetry and Image-Based Metrology	1

A.44 SCIENTIFIC WORKFLOW AND COMPUTING INFRASTRUCTURE

Domain: `workflow`. Primary tool records: 56.

ID	Subcategory	Count
workflow-scientific-workflow-orchestration	Scientific Workflow Orchestration	34
workflow-hpc-scheduling-and-distributed-execution	HPC Scheduling and Distributed Execution	21
workflow-reproducible-research-and-experiment-tracking	Reproducible Research and Experiment Tracking	7
workflow-notebooks-documents-and-scientific-reporting	Notebooks Documents and Scientific Reporting	9

A.45 GEOSPATIAL SCIENCE AND GEOGRAPHIC INFORMATION SYSTEMS

Domain: `geospatial`. Primary tool records: 53.

ID	Subcategory	Count
geospatial-desktop-gis-and-spatial-analysis	Desktop GIS and Spatial Analysis	20
geospatial-geospatial-libraries-and-coordinate-systems	Geospatial Libraries and Coordinate Systems	13
geospatial-geospatial-databases-and-web-services	Geospatial Databases and Web Services	7
geospatial-terrain-lidar-and-remote-sensing	Terrain LiDAR and Remote Sensing	17
geospatial-geomagnetic-and-geophysical-reference-computation	Geomagnetic and Geophysical Reference Computation	1
geospatial-geospatial-data-integration-and-etl	Geospatial Data Integration and ETL	1

A.46 PRODUCT LIFECYCLE MANAGEMENT

Domain: `plm`. Primary tool records: 30.

ID	Subcategory	Count
plm-product-lifecycle-and-engineering-data-management	Product Lifecycle and Engineering Data Management	22
plm-cad-documents-and-design-collaboration	CAD Documents and Design Collaboration	19
plm-bom-configuration-and-product-variability	BOM Configuration and Product Variability	9

A.47 ELECTRICAL SYSTEMS DESIGN

Domain: `electrical-design`. Primary tool records: 44.

ID	Subcategory	Count
electrical-design-electrical-schematics-and-panel-engineering	Electrical Schematics and Panel Engineering	13
electrical-design-wire-harness-and-cable-design	Wire Harness and Cable Design	10
electrical-design-electrical-installation-and-cable-sizing	Electrical Installation and Cable Sizing	9
electrical-design-grounding-lightning-and-electromagnetic-compatibility	Grounding Lightning and Electromagnetic Compatibility	14
electrical-design-electromagnetic-field-and-device-analysis	Electromagnetic Field and Device Analysis	1

A.48 SYSTEMS ENGINEERING

Domain: `systems-engineering`. Primary tool records: 83.

ID	Subcategory	Count
systems-engineering-requirements-traceability-and-verification	Requirements Traceability and Verification	25
systems-engineering-mbse-sysml-and-system-architecture	MBSE SysML and System Architecture	20
systems-engineering-system-simulation-and-co-simulation	System Simulation and Co-Simulation	28
systems-engineering-engineering-configuration-and-lifecycle-integration	Engineering Configuration and Lifecycle Integration	4
systems-engineering-formal-requirements-and-architecture-verification	Formal Requirements and Architecture Verification	4
systems-engineering-safety-reliability-and-assurance-cases	Safety Reliability and Assurance Cases	7
systems-engineering-test-automation-and-verification-management	Test Automation and Verification Management	6
systems-engineering-embedded-software-verification-and-assurance	Embedded Software Verification and Assurance	2
systems-engineering-requirements-management-and-traceability	Requirements Management and Traceability	1

A.49 RELIABILITY AND SAFETY ENGINEERING

Domain: `reliability`. Primary tool records: 40.

ID	Subcategory	Count
reliability-fault-trees-and-system-reliability-modeling	Fault Trees and System Reliability Modeling	19
reliability-failure-modes-safety-and-risk-assessment	Failure Modes Safety and Risk Assessment	17
reliability-life-data-accelerated-testing-and-reliability-prediction	Life Data Accelerated Testing and Reliability Prediction	8
reliability-maintenance-failure-reporting-and-asset-performance	Maintenance Failure Reporting and Asset Performance	5
reliability-risk-uncertainty-and-simulation-frameworks	Risk Uncertainty and Simulation Frameworks	1

A.50 METROLOGY AND DIMENSIONAL INSPECTION

Domain: metrology. Primary tool records: 137.

ID	Subcategory	Count
metrology-coordinate	Coordinate Measurement and Geometric Inspection	32
metrology-optical	Optical and Vision-Based Metrology	17
metrology-scanning	3D Scanning and Reverse Engineering	15
metrology-surface	Surface Texture and Topography	12
metrology-nde	Nondestructive Testing and Inspection	45
metrology-quality	Measurement Data, Calibration and Quality Analysis	8
metrology-virtual	Virtual Sensing and Model-Based Measurement	1
metrology-dimensional-tolerances-gd-t-and-variation-analysis	Dimensional Tolerances GD&T and Variation Analysis	5
metrology-calibration-procedures-and-instrument-asset-management	Calibration Procedures and Instrument Asset Management	9
metrology-measurement-uncertainty-and-system-capability	Measurement Uncertainty and System Capability	8
metrology-industrial-machine-vision-and-optical-inspection	Industrial Machine Vision and Optical Inspection	2
metrology-machine-tool-and-coordinate-system-calibration	Machine Tool and Coordinate System Calibration	3

A.51 INDUSTRIAL PLANT AND PIPING DESIGN

Domain: plant-design. Primary tool records: 36.

ID	Subcategory	Count
plant-design-plant-layout-piping-and-instrumentation-design	Plant Layout Piping and Instrumentation Design	11
plant-design-piping-isometrics-and-fabrication-drawings	Piping Isometrics and Fabrication Drawings	5
plant-design-pipe-stress-and-hydraulic-network-analysis	Pipe Stress and Hydraulic Network Analysis	9
plant-design-pressure-vessel-and-equipment-design	Pressure Vessel and Equipment Design	11

A.52 INFRASTRUCTURE OPERATIONS AND ASSET MANAGEMENT

Domain: infrastructure-operations. Primary tool records: 76.

ID	Subcategory	Count
infrastructure-operations-civil-infrastructure-and-asset-monitoring	Civil Infrastructure and Asset Monitoring	26
infrastructure-operations-transportation-and-traffic-simulation	Transportation and Traffic Simulation	23
infrastructure-operations-water-wastewater-and-stormwater-operations	Water Wastewater and Stormwater Operations	12
infrastructure-operations-site-design-grading-and-earthworks	Site Design Grading and Earthworks	1
infrastructure-operations-hydraulic-infrastructure-design	Hydraulic Infrastructure Design	3

ID	Subcategory	Count
infrastructure-operations- infrastructure-risk-and-incident- modeling	Infrastructure Risk and Incident Modeling	1
infrastructure-operations-road-rail- and-corridor-design	Road Rail and Corridor Design	7
infrastructure-operations-pavement- condition-and-maintenance- planning	Pavement Condition and Maintenance Plan- ning	2
infrastructure-operations- construction-machine-control- and-survey-guidance	Construction Machine Control and Survey Guidance	1
infrastructure-operations-bridge- condition-monitoring-and-asset- management	Bridge Condition Monitoring and Asset Man- agement	2
infrastructure-operations- transportation-asset-data-and- linear-referencing	Transportation Asset Data and Linear Refer- encing	1
infrastructure-operations-flood- consequences-and-evacuation- simulation	Flood Consequences and Evacuation Simula- tion	1

A.53 MANUFACTURING COST ESTIMATION

Domain: `manufacturing-cost`. Primary tool records: 27.

ID	Subcategory	Count
manufacturing-cost-manufacturing- cost-estimation-and-quoting	Manufacturing Cost Estimation and Quoting	20
manufacturing-cost-design-for- manufacturing-and-process- assessment	Design for Manufacturing and Process Assess- ment	8

A.54 CLOUD ENGINEERING AND SIMULATION PLATFORMS

Domain: `cloud-platform`. Primary tool records: 35.

ID	Subcategory	Count
cloud-platform-hpc-provisioning- and-engineering-job-execution	HPC Provisioning and Engineering Job Execu- tion	26
cloud-platform-hosted-engineering- applications-and-apis	Hosted Engineering Applications and APIs	6
cloud-platform-remote- visualization-and-engineering- desktops	Remote Visualization and Engineering Desk- tops	4
cloud-platform-scientific- software-builds-and-environment- management	Scientific Software Builds and Environment Management	2

A.55 ENGINEERING COLLABORATION AND DESIGN REVIEW

Domain: `collaboration`. Primary tool records: 20.

ID	Subcategory	Count
collaboration-engineering-model- review-and-visualization	Engineering Model Review and Visualization	11

ID	Subcategory	Count
collaboration-engineering-information-and-project-collaboration	Engineering Information and Project Collaboration	9
collaboration-technical-publishing-and-interactive-applications	Technical Publishing and Interactive Applications	5

A.56 PLASMA PHYSICS AND ENGINEERING

Domain: `plasma`. Primary tool records: 67.

ID	Subcategory	Count
plasma-pic	Particle-in-Cell and Kinetic Plasma Simulation	23
plasma-gyrokinetic	Gyrokinetic Turbulence and Transport	8
plasma-mhd	Magnetohydrodynamics and Plasma Stability	8
plasma-equilibrium	Magnetic Equilibrium and Confinement Design	12
plasma-edge	Plasma Edge, Divertors and Neutral Transport	3
plasma-discharge	Low-Temperature Plasmas and Industrial Discharges	11
plasma-integrated	Integrated Fusion and Core Transport Modeling	1
plasma-fluid	Plasma Fluid Modeling Frameworks	1
plasma-acceleration	Plasma Acceleration and Wakefields	2
plasma-electric-propulsion-and-plasma-thrusters	Electric Propulsion and Plasma Thrusters	1
plasma-energetic-particle-orbits-and-neutral-beam-injection	Energetic Particle Orbits and Neutral-Beam Injection	2
plasma-disruptions-runaway-electrons-and-plasma-startup	Disruptions Runaway Electrons and Plasma Startup	2
plasma-atomic-kinetics-and-plasma-spectroscopy	Atomic Kinetics and Plasma Spectroscopy	3
plasma-laser-pulse-preparation-and-simulation-interfaces	Laser Pulse Preparation and Simulation Interfaces	1

A.57 QUANTUM MANY-BODY AND OPEN-SYSTEM SIMULATION

Domain: `quantum-manybody`. Primary tool records: 49.

ID	Subcategory	Count
quantum-manybody-tensor-networks-and-dmrg	Tensor Networks and DMRG	19
quantum-manybody-quantum-monte-carlo-and-variational-states	Quantum Monte Carlo and Variational States	7
quantum-manybody-exact-diagonalization-and-lattice-models	Exact Diagonalization and Lattice Models	12
quantum-manybody-open-quantum-systems-and-quantum-dynamics	Open Quantum Systems and Quantum Dynamics	5
quantum-manybody-strong-correlations-and-dynamical-mean-field-theory	Strong Correlations and Dynamical Mean-Field Theory	8
quantum-manybody-superconductivity-and-mesoscopic-quantum-systems	Superconductivity and Mesoscopic Quantum Systems	1
quantum-manybody-many-body-electronic-excitations	Many-Body Electronic Excitations	1

A.58 QUANTUM COMPUTING AND CIRCUIT SIMULATION

Domain: `quantum-computing`. Primary tool records: 42.

ID	Subcategory	Count
quantum-computing-quantum-circuit-simulation-and-sdks	Quantum Circuit Simulation and SDKs	27
quantum-computing-quantum-error-correction-and-stabilizer-simulation	Quantum Error Correction and Stabilizer Simulation	4
quantum-computing-photonic-and-continuous-variable-quantum-computing	Photonic and Continuous-Variable Quantum Computing	3
quantum-computing-quantum-algorithms-and-hybrid-optimization	Quantum Algorithms and Hybrid Optimization	3
quantum-computing-quantum-hardware-control-and-pulse-modeling	Quantum Hardware Control and Pulse Modeling	2
quantum-computing-quantum-networks-and-communication	Quantum Networks and Communication	4
quantum-computing-annealing-and-binary-optimization	Annealing and Binary Optimization	2
quantum-computing-quantum-processor-layout-and-design	Quantum Processor Layout and Design	2

A.59 PARTICLE ACCELERATOR AND BEAM PHYSICS

Domain: `accelerator`. Primary tool records: 40.

ID	Subcategory	Count
accelerator-beam-optics-and-lattice-design	Beam Optics and Lattice Design	13
accelerator-particle-tracking-and-beam-dynamics	Particle Tracking and Beam Dynamics	29
accelerator-accelerator-electromagnetics-and-wakefields	Accelerator Electromagnetics and Wakefields	7
accelerator-charged-particle-optics-and-sources	Charged-Particle Optics and Sources	4
accelerator-synchrotron-radiation-and-beamline-engineering	Synchrotron Radiation and Beamline Engineering	1

A.60 FIRE AND PROCESS SAFETY ENGINEERING

Domain: `fire-safety`. Primary tool records: 27.

ID	Subcategory	Count
fire-safety-chemical-release-consequences-and-quantitative-risk	Chemical Release Consequences and Quantitative Risk	6
fire-safety-wildland-fire-behavior-and-atmosphere-coupling	Wildland Fire Behavior and Atmosphere Coupling	6
fire-safety-fire-protection-design-and-code-compliance	Fire Protection Design and Code Compliance	1
fire-safety-fire-smoke-and-ventilation-simulation	Fire Smoke and Ventilation Simulation	2
fire-safety-pedestrian-movement-and-evacuation	Pedestrian Movement and Evacuation	4
fire-safety-structural-response-to-fire	Structural Response to Fire	4

ID	Subcategory	Count
fire-safety-sprinkler-design-and-hydraulic-calculation	Sprinkler Design and Hydraulic Calculation	3
fire-safety-smoke-control-and-building-airflow	Smoke Control and Building Airflow	1

A.61 PLANETARY SCIENCE

Domain: planetary-science. Primary tool records: 14.

ID	Subcategory	Count
planetary-science-planetary-atmosphere-spectra-and-retrieval	Planetary Atmosphere Spectra and Retrieval	4
planetary-science-surface-chronology-and-crater-analysis	Surface Chronology and Crater Analysis	1
planetary-science-planetary-climate-and-general-circulation	Planetary Climate and General Circulation	3
planetary-science-planetary-imaging-mapping-and-mission-data	Planetary Imaging Mapping and Mission Data	2
planetary-science-engineering-atmosphere-and-environment-models	Engineering Atmosphere and Environment Models	2
planetary-science-planetary-surface-thermal-modeling	Planetary Surface Thermal Modeling	1
planetary-science-impact-cratering-and-shock-physics	Impact Cratering and Shock Physics	1

A.62 SOLID EARTH SCIENCES

Domain: geoscience. Primary tool records: 58.

ID	Subcategory	Count
geoscience-landscape-evolution-and-sedimentary-basins	Landscape Evolution and Sedimentary Basins	4
geoscience-mineral-thermodynamics-and-petrology	Mineral Thermodynamics and Petrology	5
geoscience-crustal-deformation-and-earthquake-mechanics	Crustal Deformation and Earthquake Mechanics	2
geoscience-geological-modeling-and-subsurface-interpretation	Geological Modeling and Subsurface Interpretation	6
geoscience-geochronology-and-paleoclimate-age-models	Geochronology and Paleoclimate Age Models	5
geoscience-gnss-geodesy-and-tectonic-positioning	GNSS Geodesy and Tectonic Positioning	4
geoscience-mantle-convection-and-lithosphere-dynamics	Mantle Convection and Lithosphere Dynamics	5
geoscience-structural-geology-and-orientation-analysis	Structural Geology and Orientation Analysis	3
geoscience-plate-reconstruction-and-tectonic-evolution	Plate Reconstruction and Tectonic Evolution	3
geoscience-geophysical-inversion-and-potential-fields	Geophysical Inversion and Potential Fields	4
geoscience-seismic-data-processing-and-monitoring	Seismic Data Processing and Monitoring	6
geoscience-earthquake-location-hazard-and-risk	Earthquake Location Hazard and Risk	3

ID	Subcategory	Count
geoscience-seismic-wave-propagation-and-dynamic-rupture	Seismic Wave Propagation and Dynamic Rupture	3
geoscience-geodynamics-modeling-frameworks	Geodynamics Modeling Frameworks	1
geoscience-geostatistics-and-spatial-property-modeling	Geostatistics and Spatial Property Modeling	3
geoscience-geochronology-and-geochemical-data-analysis	Geochronology and Geochemical Data Analysis	1

A.63 MICROFLUIDIC ENGINEERING

Domain: `microfluidics`. Primary tool records: 7.

ID	Subcategory	Count
microfluidics-microfluidic-device-design-and-automation	Microfluidic Device Design and Automation	5
microfluidics-microscale-flow-transport-and-electrokinetics	Microscale Flow Transport and Electrokinetics	2

A.64 DATA CENTER ENGINEERING

Domain: `data-center`. Primary tool records: 12.

ID	Subcategory	Count
data-center-cooling-airflow-and-thermal-design	Cooling Airflow and Thermal Design	4
data-center-infrastructure-capacity-layout-and-asset-planning	Infrastructure Capacity Layout and Asset Planning	6
data-center-operational-thermal-and-power-optimization	Operational Thermal and Power Optimization	1
data-center-workload-scheduling-and-energy-simulation	Workload Scheduling and Energy Simulation	1

A.65 AGRICULTURAL ENGINEERING

Domain: `agricultural`. Primary tool records: 5.

ID	Subcategory	Count
agricultural-greenhouse-climate-and-energy-engineering	Greenhouse Climate and Energy Engineering	1
agricultural-agricultural-application-and-spray-engineering	Agricultural Application and Spray Engineering	1
agricultural-irrigation-design-and-hydraulics	Irrigation Design and Hydraulics	3

A.66 NAVAL ARCHITECTURE AND MARINE ENGINEERING

Domain: `marine`. Primary tool records: 73.

ID	Subcategory	Count
marine-seakeeping-wave-loads-and-potential-flow-hydrodynamics	Seakeeping Wave Loads and Potential-Flow Hydrodynamics	16
marine-mooring-risers-cables-and-offshore-dynamics	Mooring Risers Cables and Offshore Dynamics	6

ID	Subcategory	Count
marine-marine-operations-and-coupled-vessel-dynamics	Marine Operations and Coupled Vessel Dynamics	9
marine-vessel-resistance-propellers-and-propulsion	Vessel Resistance Propellers and Propulsion	12
marine-hydrodynamic-model-preparation-and-data-exchange	Hydrodynamic Model Preparation and Data Exchange	2
marine-hull-structural-design-and-classification-rules	Hull Structural Design and Classification Rules	8
marine-integrated-ship-design-and-production-engineering	Integrated Ship Design and Production Engineering	8
marine-hull-form-design-and-fairing	Hull Form Design and Fairing	7
marine-hydrostatics-stability-and-loading	Hydrostatics Stability and Loading	11
marine-viscous-flow-and-marine-cfd	Viscous Flow and Marine CFD	4

A.67 MINING AND MINERAL PROCESSING

Domain: mining. Primary tool records: 32.

ID	Subcategory	Count
mining-drill-and-blast-design-and-analysis	Drill-and-Blast Design and Analysis	6
mining-mineral-processing-and-metallurgical-flowsheets	Mineral Processing and Metallurgical Flowsheets	9
mining-mine-design-production-planning-and-scheduling	Mine Design Production Planning and Scheduling	8
mining-geological-models-and-mineral-resource-estimation	Geological Models and Mineral Resource Estimation	4
mining-haulage-fleet-and-mine-logistics-simulation	Haulage Fleet and Mine Logistics Simulation	1
mining-exploration-drillhole-and-assay-data-management	Exploration Drillhole and Assay Data Management	1
mining-mine-ventilation-cooling-and-environmental-control	Mine Ventilation Cooling and Environmental Control	3

A.68 URBAN AND REGIONAL PLANNING

Domain: urban-planning. Primary tool records: 16.

ID	Subcategory	Count
urban-planning-urban-massing-zoning-and-site-feasibility	Urban Massing Zoning and Site Feasibility	4
urban-planning-semantic-city-models-and-geospatial-integration	Semantic City Models and Geospatial Integration	2
urban-planning-land-use-scenarios-and-spatial-suitability	Land-Use Scenarios and Spatial Suitability	1
urban-planning-urban-microclimate-and-environmental-simulation	Urban Microclimate and Environmental Simulation	6
urban-planning-urban-energy-and-infrastructure-planning	Urban Energy and Infrastructure Planning	1
urban-planning-urban-land-use-and-development-microsimulation	Urban Land-Use and Development Microsimulation	2

A.69 MECHANICAL SYSTEMS AND MACHINE ELEMENTS

Domain: mechanical-systems. Primary tool records: 70.

ID	Subcategory	Count
mechanical-systems-bearings-lubrication-and-rolling-contacts	Bearings Lubrication and Rolling Contacts	16
mechanical-systems-powertrain-dynamics-durability-and-nvh	Powertrain Dynamics Durability and NVH	7
mechanical-systems-contact-mechanics-friction-and-wear	Contact Mechanics Friction and Wear	5
mechanical-systems-gears-transmissions-and-machine-element-design	Gears Transmissions and Machine-Element Design	33
mechanical-systems-mechanical-seal-simulation	Mechanical Seal Simulation	2
mechanical-systems-turbomachinery-meanline-and-throughflow-design	Turbomachinery Meanline and Throughflow Design	3
mechanical-systems-shafts-bearings-and-drivetrain-analysis	Shafts Bearings and Drivetrain Analysis	6
mechanical-systems-gear-metrology-and-manufacturing-verification	Gear Metrology and Manufacturing Verification	2

A.70 ATMOSPHERIC SCIENCE AND METEOROLOGY

Domain: atmospheric-science. Primary tool records: 65.

ID	Subcategory	Count
water-atmosphere	Atmospheric Composition and Air Quality	26
water-weather	Weather Prediction and Climate Modeling	21
atmospheric-science-weather-and-climate-data-processing	Weather and Climate Data Processing	4
atmospheric-science-atmospheric-radiative-transfer	Atmospheric Radiative Transfer	7
atmospheric-science-earth-system-model-coupling-and-infrastructure	Earth-System Model Coupling and Infrastructure	1
atmospheric-science-earth-system-data-assimilation	Earth-System Data Assimilation	4
atmospheric-science-forecast-verification-and-evaluation	Forecast Verification and Evaluation	2

A.71 EMBEDDED SOFTWARE ENGINEERING

Domain: embedded-software. Primary tool records: 68.

ID	Subcategory	Count
embedded-software-embedded-ides-and-development-toolchains	Embedded IDEs and Development Toolchains	12
embedded-software-compilers-and-cross-compilation	Compilers and Cross Compilation	4
embedded-software-real-time-operating-systems-and-partitioning	Real-Time Operating Systems and Partitioning	12
embedded-software-autosar-architecture-and-basic-software	AUTOSAR Architecture and Basic Software	7
embedded-software-virtual-ecus-and-embedded-system-simulation	Virtual ECUs and Embedded System Simulation	7
embedded-software-ecu-measurement-calibration-and-diagnostics	ECU Measurement Calibration and Diagnostics	2

	ID	Subcategory	Count
1782			
1783	embedded-software-model-based-	Model-Based Embedded Code Generation	2
1784	embedded-code-generation		
1785	embedded-software-static-analysis-	Static Analysis and Software Verification	9
1786	and-software-verification		
1787	embedded-software-embedded-	Embedded Debugging Trace and Profiling	4
1788	debugging-trace-and-profiling		
1789	embedded-software-embedded-	Embedded Linux Distribution and Image	2
1790	linux-distribution-and-image-builds	Builds	
1791	embedded-software-embedded-unit-	Embedded Unit Integration and Regression	2
1792	integration-and-regression-testing	Testing	
1793	embedded-software-build-systems-	Build Systems and Dependency Management	5
1794	and-dependency-management		
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